

Fig. 4: Actual-size, single-side PCB for the phase angle control of SCR using AT89C51

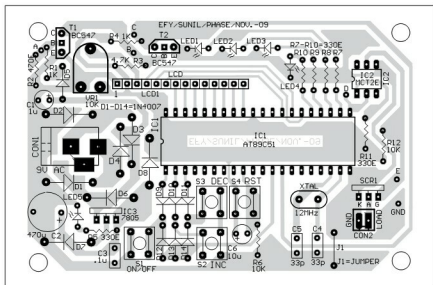


Fig. 5: Component layout for the PCB

waveform.

4. This negative pulse is fed to the interrupt pin (INT1) of the microcontroller, which acts as an interrupt for the microcontroller. The microcontroller then generates a positive pulse on P2.0 (at point 'D') after some delay. This turns 'off' the internal LED of the opto-coupler (MCT2E) and a positive pulse is produced at output 'E'. This is used for triggering (fire) SCR1.

5. Depending on the time delay in between the interrupt and the pulse on port pin P2.0 of the microcontroller, the

SCR is fired in the middle of the half-wave cycle.

6. Two different waveforms—one for 4 ms delay and the other for 8 ms delay—are shown in Fig. 3. In the case of 4 ms delay, the output positive cycle of the AC wave is 60 per cent of the input. Therefore, nearly 60 per cent of the power is delivered to the load (the dotted line shows part of waveform that has been cut). In the second case of 8 ms delay, the output cycle is 20 per cent of the input cycle, so only 20 per cent of the power is delivered to the load.

TABLE II
Change in Delay Using
Push-to-on Switches

LED	Indication
LED1	SCR on
LED2	SCR off
LED3	Blinks when delay is increased
LED4	Blinks when delay is decreased

This change in delay is done using switches S1 and S2. Different LEDs are used for indicating different functions as shown in Table II.

The diodes D12 through D14 are connected in such a manner that whenever any of the three push-to-on switches are pressed, it generates an external interrupt $\overline{\text{INT0}}$.

When switch S1 is pressed for the first time, it enables external interrupt INT1 and displays the message 'SCR on.' So after every 10 ms, external interrupt INT1 is generated which starts the entire operation. Pressing switch S1 again disables external interrupt 0 and the message 'SCR off' is displayed. The complete SCR operation gets shut off.

On pressing S2, the delay increases by 1 ms (firing angle will shift by 18°) and firing of SCR is delayed by 1 ms. The power delivered to the load is also decreased by 10 per cent. The maximum delay that can be applied is 9 ms which will delay firing by an angle of 162° . When the limit is reached, it is indicated by LED3 and a message 'Max. phase angle' is displayed on the LCD. The glowing of the bulb goes off.

Similarly, when S3 is pressed, the delay is decreased by 1 ms and the load current increases by 10 per cent. The minimum delay is 0 ms, which means a full positive cycle is applied. However, when the limit is reached, it is indicated by LED4 and a message 'Min. phase angle' is displayed.

An actual-size, single-side PCB for phase-angle control using SCR is shown in Fig. 4 and its component layout in Fig. 5.